Maximum Principle for Absolute Minimisers

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Abstract

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1 Introduction

For $n, N, k \in \mathbb{N}$, let $\Omega \subset \mathbb{R}^n$ be an open bounded set and $H : \Omega \times \mathbb{R}^N \times \mathbb{R}^{Nn} \times \ldots \times \mathbb{R}^{Nn^k} \to \mathbb{R}$ be $\mathscr{L}(\Omega) \otimes \mathscr{B}(\mathbb{R}^\alpha)$-measurable, where we have identified $\mathbb{R}^\alpha \cong \mathbb{R}^N \times \mathbb{R}^{Nn} \times \ldots \times \mathbb{R}^{Nn^k}$ for $\alpha = \alpha(n, N, k) = N\frac{n^{k+1}-1}{n-1}$. Here $\mathscr{L}(\Omega)$ is the Lebesgue σ-algebra of Ω and $\mathscr{B}(\mathbb{R}^\alpha)$ is the Borel σ-algebra of $\mathbb{R}^\alpha$. Let $U \subseteq \Omega$ be an open subset and consider the supremal functional

$$E_\infty(u, U) := \operatorname*{ess\,sup}_{x \in U} H\big(x, \mathrm{D}^{[k]}u(x)\big), \tag{1.1}$$

where $u \in W^{k,\infty}(\Omega, \mathbb{R}^N)$ and $\mathrm{D}^{[k]}u := (u, \mathrm{D}u, \ldots, \mathrm{D}^k u)$ is the jet of order k of u.
The goal herein is to prove a maximum principle property for the absolute minimisers of the functional E_∞. We recall that $u \in W^{k,\infty}(\Omega, \mathbb{R}^N)$ is an absolute minimiser of E_∞ if

$$E_\infty(u, U) \leq E_\infty(u + \Phi, U) \qquad \forall \Phi \in W_0^{k,\infty}(U, \mathbb{R}^N) \quad \forall U \subseteq \Omega \text{ open.}$$

Under very mild assumptions on the supremand H, we prove that for an absolute minimiser the essential supremum in (1.1) is "attained at the boundary of U". In order to give a precise meaning to this expression, we refer to the supremum on ∂U of the L^∞-representative of $H(\cdot, \mathrm{D}^{[k]}u)$, which is defined as

$$H(x, \mathrm{D}^{[k]}u(x))^* := \lim_{\varepsilon \to 0} \operatorname*{ess\,sup}_{B_\varepsilon(x) \cap U} H(\cdot, \mathrm{D}^{[k]}u) \qquad \forall x \in \partial U. \tag{1.2}$$

Therefore, we say that $u \in W^{k,\infty}(\Omega, \mathbb{R}^N)$ satisfies the maximum principle for E_∞ in Ω if

$$\operatorname*{ess\,sup}_U H(\cdot, \mathrm{D}^{[k]}u) = \sup_{\partial U} H(\cdot, \mathrm{D}^{[k]}u)^* \qquad \forall U \subseteq \Omega \text{ open.}$$

Notice that this property is only necessary for absolute minimisers: the cone function $u(x) = |x|$ belongs to $W^{1,\infty}(\mathbb{R}^n)$ and clearly satisfies the maximum principle for $E_\infty(u, \Omega) := \operatorname{ess\,sup}_\Omega |\mathrm{D}u|$, but it is not an absolute minimiser. However, we will see that the maximum principle is a property that characterises a class of maps which is larger than the one of absolute minimisers, and are the ones that minimise E_∞ with respect to compactly supported variations.

Before stating our main results, we underline that E_∞ in (1.1) is the general object of study in the L^∞-Calculus of Variations, a field initiated by Aronsson in [3]. His pioneering work on the scalar first order case (namely when $N = k = 1$) has been well developed by now, and most challenges have been thoroughly analysed and understood (see [18] for a survey reference). More recently, in [17], the second author approached the vectorial case ($N > 1$), which turned out to be more involved and intriguing than the scalar one. For this reason, a complete theory is still far from reach. The situation is very similar in the higher order case ($k > 1$), recently started in [21, 26]. Without any pretension of being exhaustive, we refer also to [17, 19, 20, 22] for a glimpse of the literature on vectorial first order problems and to [9, 13, 23–25] for higher order ones. Furthermore, the fractional order case ($k \notin \mathbb{N}$) has been recently explored in [10].

It is worth to mention that, given boundary data $u_0 \in W^{k,\infty}(\Omega, \mathbb{R}^N)$, the existence (and/or uniqueness) of absolute minimisers to the Dirichlet problem in $W^{k,\infty}_{u_0}(\Omega, \mathbb{R}^N)$ associated to E_∞ in (1.1) is an open problem already when $k = 1$ and $n, N > 1$ or when $N = 1$ and $n, k > 1$. There are some exceptions in the second order case, in case H has special dependence in the second derivatives of u. We refer to [21, 24, 26] for the scalar second order case and to [9] for the vectorial one (see also [10] for the fractional case).

Now we proceed to specify the assumptions on the function H. By denoting $X = (X_0, X_1, \dots, X_k) \in R^N \times \mathbb{R}^{Nn} \times \dots \mathbb{R}^{Nn^k} = \mathbb{R}^\alpha$, we assume that $H : \Omega \times \mathbb{R}^\alpha \to \mathbb{R}$ is Carathéodory and radially non-decreasing w.r.t. X, i.e. for almost every $x \in \Omega$ and every $X \in \mathbb{R}^\alpha$ the function

$$t \mapsto H(x, tX) \text{ is non-decreasing in } \mathbb{R}^+.$$

Notice that radial monotonicity implies that, for almost every $x \in \Omega$, $H(x, \cdot)$ has a global minimum at $X = 0$. Without loss of generality we can assume that $H(x, 0) = 0$ a.e. $x \in \Omega$, so that H takes values in $\mathbb{R}^+$ and we can write $E_\infty(u, U) = \|H(\cdot, \mathrm{D}^{[k]}u)\|_{L^\infty(U)}$ for every $U \subseteq \Omega$ open.

We point out that any level convex function with global minimum at the origin is necessarily radially non-decreasing, so that our result applies to the class of level convex functions, which is a common assumption for H in L^∞ variational problems (see e.g. [4]). Moreover, one can even apply the result to non-continuous supremands: indeed, for a function $g : \mathbb{R}^+ \to \mathbb{R}^+$ lower semicontinuous and non-decreasing, set $G = g \circ H$ and $E'_\infty(u, U) := \operatorname{ess\,sup}_U G(\cdot, \mathrm{D}^{[k]}u)$. It is not difficult to see that E_∞ and E'_∞ share the same minimisers. In addition, if $u \in W^{k,\infty}(\Omega, \mathbb{R}^N)$ satisfies the maximum principle for E_∞ in Ω, then for every $U \subseteq \Omega$ open

$$E'_\infty(u, U) = g\left(\operatorname*{ess\,sup}_U H(\cdot, \mathrm{D}^{[k]}u)\right) = g\left(\operatorname*{ess\,sup}_{\partial U} H(\cdot, \mathrm{D}^{[k]}u)^*\right) = \left(\operatorname*{ess\,sup}_{\partial U} G(\cdot, \mathrm{D}^{[k]}u)^*\right),$$

i.e. u satisfies the maximum principle for E'_∞ as well.

Now we can state our main result.

Theorem 1.1 (Maximum Principle for Absolute Minimisers)**.** *Let $\Omega \subset \mathbb{R}^n$ be a bounded open set and let $H : \Omega \times \mathbb{R}^\alpha \to \mathbb{R}^+$ be a Carathéodory function with $H(x, 0) = 0$ and $H(x, \cdot)$ radially non-decreasing for a.e. $x \in \Omega$. Assume that $u \in W^{k,\infty}(\Omega, \mathbb{R}^N)$ is an absolute minimiser for the functional E_∞, defined in* (1.1)*. Then u satisfies the maximum principle for E_∞ in Ω, namely*

$$\operatorname*{ess\,sup}_U H(\cdot, \mathrm{D}^{[k]}u) = \sup_{\partial U} H(\cdot, \mathrm{D}^{[k]}u)^* \qquad \forall U \subseteq \Omega \text{ open.} \tag{1.3}$$

where $H(\cdot, \mathrm{D}^{[k]}u)^*$ is the L^∞-representative of $H(\cdot, \mathrm{D}^{[k]}u)$, defined in (1.2)*.*

Before stating our second theorem, we give the definition of absolute minimiser w.r.t. compactly supported variations.

Definition 1.2 (The class $\mathrm{AM}_c(\Omega, \mathbb{R}^N)$)**.** *We say that $u \in W^{k,\infty}(\Omega, \mathbb{R}^N)$ is an absolute minimiser w.r.t. compactly supported variations, and we write $u \in \mathrm{AM}_c(\Omega, \mathbb{R}^N)$, if*

$$E_\infty(u, U) \leq E_\infty(u + \Phi, U) \qquad \forall \Phi \in W_c^{k,\infty}(\Omega, \mathbb{R}^N) \quad \forall U \subseteq \Omega \text{ open.}$$

Our second result is the following.

Theorem 1.3. *Let $\Omega \subset \mathbb{R}^n$ be a bounded open set and let $H : \Omega \times \mathbb{R}^\alpha \to \mathbb{R}^+$ be a Carathéodory function with $H(x, 0) = 0$ and $H(x, \cdot)$ radially non-decreasing for a.e. $x \in \Omega$. Consider the functional E_∞, defined in* (1.1)*. Then, the following are equivalent for a map $u \in W^{k,\infty}(\Omega, \mathbb{R}^N)$:*

i) u satisfies the maximum principle for E_∞ in Ω;

ii) $u \in \mathrm{AM}_c(\Omega, \mathbb{R}^N)$.

The proof of Theorems 1.1 and 1.3 are in Sections ...., after a short preliminary section.

2 Preparatory tools

2.1 On radial monotonicity

In this first preliminary section, we recall some notions of radial monotonicity for function on $\mathbb{R}^\alpha$, together with some relevant examples.

Definition 2.1. *A function $H : \mathbb{R}^\alpha \to \mathbb{R}$ is said to be*

i) radially non-decreasing if for every $X \in \mathbb{R}^\alpha$ the function $t \mapsto H(tX)$ is non-decreasing in $\mathbb{R}^+$;

ii) strictly radially increasing if for every $X \in \mathbb{R}^\alpha$ the function $t \mapsto H(tX)$ is strictly increasing in $\mathbb{R}^+$;

iii) strongly radially increasing if there exists a continuous function $c : [0, 1] \times \mathbb{R}^+ \to \mathbb{R}^+$, with $c(1, \cdot) = c(\cdot, 0) = 0$ and $c > 0$ in $(0, 1) \times (0, \infty)$, such that

$$H(tX) \leq H(X) - c(t, |X|) \qquad \forall t \in [0, 1] \quad \forall X \in \mathbb{R}^\alpha.$$

Notice that *i)* $\Rightarrow$ *ii)* $\Rightarrow$ *iii)*, and implications do not invert in general. The above definition is in complete analogy with the notion of level convexity and its stronger versions. In particular, any level convex (resp. strictly level convex) function with global minimum at the origin is radially non-decreasing (resp. strictly radially increasing). Of course, a radially non-decreasing function need not to be level convex, since its level sets need only to be star shaped w.r.t. the origin.

Strongly radially increasing functions can be seen as strict radially increasing functions with a modulus of monotonicity. This class of functions will be particularly important for us, since in the proof of Theorem 1.1 we will use it to approximate radially non-decreasing functions in the uniform convergence.

Example 2.2. Relevant examples of strongly radially increasing functions are:

- Any strongly convex function with global minimum at the origin;
- The cone function associated to a (smooth) open set $S \subset \mathbb{R}^\alpha$ star shaper w.r.t. the origin with $\nu_S(X) \cdot X > 0$ for every $X \in \partial S$, where ν_S is the outward normal vector to ∂S.
- Functions of the form $H_{\lambda,\beta}(X) = H(X) + \lambda |X|^\beta$, for $\lambda, \beta > 0$ and H radially non-decreasing.

2.2 Properties of convex functions

In this second preliminary section, we prove some results concerning convex functions on $\mathbb{R}^N$. In what follows, we denote by F_ξ and $F_{\xi\xi}$ respectively the gradient and the Hessian of a function $F : \mathbb{R}^N \to \mathbb{R}$.

Lemma 2.3. *Let $F : \mathbb{R}^N \to \mathbb{R}$ be C^2 and convex, with $F(0) = 0$. Then*

$$F(\xi) \leq F_\xi(\xi) \cdot \xi \quad \forall \xi \in \mathbb{R}^N.$$

Proof. For $\xi \in \mathbb{R}^N$, set $\Phi(\xi) := F_\xi(\xi) \cdot \xi$. Then $\Phi_\xi(\xi) = F_{\xi\xi}(\xi)\xi + F_\xi(\xi)$. For any $\lambda \geq 0$, since F is convex, we get

$$\frac{d}{d\lambda}[\Phi(\lambda\xi)] = \Phi_\xi(\lambda\xi) \cdot \xi = \lambda F_{\xi\xi}(\lambda\xi)\xi \cdot \xi + F_\xi(\lambda\xi) \cdot \xi \geq F_\xi(\lambda\xi) \cdot \xi = \frac{d}{d\lambda}[F(\lambda\xi)].$$

Therefore, by integrating on $(0, 1)$, we conclude by using $F(0) = 0$, as

$$\Phi(\xi) = \int_0^1 \frac{d}{d\lambda}[\Phi(\lambda\xi)]d\lambda \geq \int_0^1 \frac{d}{d\lambda}[F(\lambda\xi)]d\lambda = F(\xi).$$

□

We also need the next invertibility result.

Lemma 2.4. *Let $F : \mathbb{R}^N \to \mathbb{R}$ be C^2 and strictly convex, with $F \geq 0$ and $F(0) = 0$. Then the map $\Phi : \mathbb{R}^N \to \mathbb{R}^N$ defined by*

$$\Phi(\xi) := F(\xi)\frac{F_\xi(\xi)}{|F_\xi(\xi)|} \quad \forall \xi \in \mathbb{R}^N$$

is a homeomorphism in $\mathbb{R}^N$.

Proof. We divide the proof in two steps, showing first that Φ is a local diffeomorphism in $\mathbb{R}^N \setminus \{0\}$, and secondly that it is a global homeomorphism in $\mathbb{R}^N$.
Step 1: We show that $\det\Phi_\xi(\xi) > 0$ away from the origin.
Let us compute the derivative of Φ at fixed $\xi \neq 0$:

$$\begin{aligned}\Phi_\xi &= \frac{1}{|F_\xi|}\left(F_\xi \otimes F_\xi + FF_{\xi\xi}\left(I - \frac{F_\xi \otimes F_\xi}{|F_\xi|^2}\right)\right) \\ &= \frac{1}{|F_\xi|}\left(FF_{\xi\xi} + (|F_\xi|^2 I - FF_{\xi\xi})\frac{F_\xi \otimes F_\xi}{|F_\xi|^2}\right), \quad \text{in } \mathbb{R}^N \setminus \{0\}.\end{aligned}$$

Notice that it is well defined and continuous in $\mathbb{R}^N \setminus \{0\}$, since our assumptions imply that both F and F_ξ vanish only at $\xi = 0$. Now we recall the following formula concerning the determinant of rank 1-perturbation of invertible matrices:

$$\det(A + a \otimes b) = (\det A)(1 + (A^{-1}a) \cdot b) \quad \forall A \in GL_N(\mathbb{R}) \quad \forall a, b \in \mathbb{R}^N.$$

Applying it to $A = FF_{\xi\xi}$, $a = (|F_\xi|^2 I - FF_{\xi\xi})\frac{F_\xi}{|F_\xi|}$, $b = \frac{F_\xi}{|F_\xi|}$, we get

$$\begin{aligned}\det\Phi_\xi &= \frac{1}{|F_\xi|^N}\det(FF_{\xi\xi})\left(1 + (FF_{\xi\xi})^{-1}\left((|F_\xi|^2 I - FF_{\xi\xi})\frac{F_\xi}{|F_\xi|}\right) \cdot \frac{F_\xi}{|F_\xi|}\right) \\ &= \frac{F^{N-1}}{|F_\xi|^N}\det(F_{\xi\xi})\left((F_{\xi\xi})^{-1}F_\xi \cdot F_\xi\right) \\ &> 0 \quad \text{in } \mathbb{R}^N \setminus \{0\},\end{aligned}$$

where we used that $\det(F_{\xi\xi}) > 0$ and that $(F_{\xi\xi})^{-1}$ is positive definite, by strict convexity of F.

Step 2: We show that Φ is a global homeomorphism in $\mathbb{R}^N$.
Notice that Φ is continuous on $\mathbb{R}^N$ and $\Phi(0) = 0$. Since F is strictly convex, positive and null at 0, we have that $\Phi^{-1}(\{0\}) = \{0\}$, which implies $\Phi(\mathbb{R}^N \setminus \{0\}) \subset \mathbb{R}^N \setminus \{0\}$. Let us prove that the restriction $\Phi|_{\mathbb{R}^N\setminus\{0\}} : \mathbb{R}^N \setminus \{0\} \to \mathbb{R}^N \setminus \{0\}$ is surjective, by showing that $\Phi(\mathbb{R}^N \setminus \{0\})$ is open and closed in $\mathbb{R}^N \setminus \{0\}$.
Since Φ is a local diffeomorphism in $\mathbb{R}^N \setminus \{0\}$, it is immediate to prove that $\Phi(\mathbb{R}^N \setminus \{0\})$ is open. To show the closedness, consider a sequence $(\eta_k)_k \subset \Phi(\mathbb{R}^N \setminus \{0\})$ converging to some $\eta_\infty \in \mathbb{R}^N \setminus \{0\}$ and let $(\xi_k)_k \in \mathbb{R}^N \setminus \{0\}$ be such that $\Phi(\xi_k) = \eta_k$. Our assumptions on F implies

$$\lim_{|\xi|\to\infty} F(\xi) = \lim_{|\xi|\to\infty} |\Phi(\xi)| = \infty. \tag{2.1}$$

So, $(\xi_k)_k$ is necessarily bounded and there exists a subsequence $(\xi_{k_j})_j \subset (\xi_k)_k$ such that $\xi_{k_j} \to \xi_\infty$, for some $\xi_\infty \in \mathbb{R}^N$, as $j \to \infty$. Therefore, we conclude

$$\eta_\infty = \lim_{j\to\infty} \eta_{k_j} = \lim_{j\to\infty} \Phi(\xi_{k_j}) = \Phi(\xi_\infty),$$

that also ensures $\xi_\infty \neq 0$.
Now we claim that any $\eta \in \mathbb{R}^N \setminus \{0\}$ has finitely many preimages. Indeed, if for the sake of contradiction we assume there exists a sequence of distinct points $(\xi_k)_k \subset \mathbb{R}^N$ such that $\Phi(\xi_k) = \eta$, then $(\xi_k)_k$ is bounded, from (2.1). So, there exists a subsequence $(\xi_{k_j})_j \subset (\xi_k)_k$ such that $\xi_{k_j} \to \xi_\infty$, for some $\xi_\infty \in \mathbb{R}^N \setminus \{0\}$, as $j \to \infty$. Then any neighbourhood of ξ_∞ contains infinitely many preimages of η, contradicting the local invertibility of Φ in $\mathbb{R}^N \setminus \{0\}$. Therefore, we have proved that Φ is a topological covering of $\mathbb{R}^N \setminus \{0\}$. For $N \geq 3$, we infer that Φ is a homeomorphism in $\mathbb{R}^N \setminus \{0\}$, since it is a universal covering of a simply connected space (see [28, Prop. 13.30]). Finally, since Φ is continuous at 0 and $\Phi^{-1}(0) = \{0\}$, it is not difficult to see that Φ extends to a global homeomorphism in $\mathbb{R}^N$, for any $N \geq 3$.
It remains to discuss the case $N \leq 2$. The one dimensional case is immediate, so let us assume $N = 2$. Extend F to $\mathbb{R}^3$ by defining the function

$$\tilde{F}(\xi_1, \xi_2, \xi_3) = F(\xi_1, \xi_2) + \xi_3^2.$$

Then $\tilde{F} : \mathbb{R}^3 \to \mathbb{R}$ is C^2 and strictly convex, with $\tilde{F} \geq 0$ and $\tilde{F}(0) = 0$. Thanks to the previuos argument, the map

$$\tilde{\Phi}(\xi) = \tilde{F}(\xi)\frac{\tilde{F}_\xi(\xi)}{|\tilde{F}_\xi(\xi)|} \quad \forall \xi \in \mathbb{R}^3$$

is a homeomorphism in $\mathbb{R}^3$. Since $\tilde{\Phi}|_{\mathbb{R}^2\times\{0\}} = \Phi$, we obtain the conclusion. □

3 Proof of the main results

We divide the proof of Theorem **??** into three parts, concerning existence, PDE derivation and uniqueness respectively. In the first two, our arguments follow similar lines to those in [21], appropriately extended to the vectorial setting up to some technical issues, while in the last part our arguments are significantly different.

3.1 Existence

Let $p \geq 1$ and recall that $u_0 \in \mathcal{W}^{2,\infty}(\Omega, \mathbb{R}^N)$. We begin by considering the functional

$$E_p(u) := \left(\fint_\Omega F(x, \mathrm{L}u(x))^p dx \right)^{\frac{1}{p}}, \quad u \in W^{2,2p}_{u_0}(\Omega, \mathbb{R}^N).$$

Assume that $E_p(u) < \infty$ for some $u \in W^{2,2p}_{u_0}(\Omega, \mathbb{R}^N)$. By making use of (**??**) and Corollary **??**, we have $\mathrm{L}u \in L^{2p}(\Omega, \mathbb{R}^N)$ and

$$\|u\|_{2,2p} \leq C \left(\|\mathrm{L}u\|_{2p} + \|u_0\|_{2,2p} \right) \leq C \left(c^{\frac{1}{2}} |\Omega|^{\frac{1}{2p}} E_p(u)^{\frac{1}{2}} + \|u_0\|_{2,2p} \right)$$

for a constant $C = C(\Omega, p, \lambda, A)$. Thus, we deduce that E_p is coercive in $W^{2,2p}_{u_0}(\Omega, \mathbb{R}^N)$. Moreover, since F is convex with respect to ξ, by Direct Methods we obtain the existence of a minimiser $u_p \in W^{2,2p}_{u_0}(\Omega, \mathbb{R}^N)$ of E_p.
Fix $k \in \mathbb{N}$. Extend E_p and E_∞ to $W^{2,k}_{u_0}(\Omega, \mathbb{R}^N)$ as

$$E_p(u) = \begin{cases} \left(\fint_\Omega F(x, \mathrm{L}u(x))^p \right)^{\frac{1}{p}}, & u \in W^{2,2p}_{u_0}, \\ +\infty &, \quad u \in W^{2,k}_{u_0} \setminus W^{2,2p}_{u_0}, \end{cases} \qquad E_\infty(u) = \begin{cases} \|F(\cdot, \mathrm{L}u)\|_\infty, & u \in \mathcal{W}^{2,\infty}_{u_0}, \\ +\infty &, \quad u \in W^{2,k}_{u_0} \setminus \mathcal{W}^{2,\infty}_{u_0}. \end{cases}$$

Let us show that $(E_p)_p$ is equi-coercive in $W^{2,k}_{u_0}$ with respect to p. Assume $E_p(u) \leq M$ for all $p \in [k/2, \infty)$. By (**??**) and the Hölder inequality, we have

$$c^{-\frac{1}{2}} |\Omega|^{-\frac{1}{2k}} \|\mathrm{L}u\|_k \leq E_k(u)^{\frac{1}{2}} \leq E_{2p}(u)^{\frac{1}{2}} \leq M^{\frac{1}{2}}. \tag{3.1}$$

Hence, by Corollary **??**

$$\|u\|_{2,k} \leq C \left(\|\mathrm{L}u\|_k + \|u_0\|_{2,k} \right) \leq C \left(c^{\frac{1}{2}} |\Omega|^{\frac{1}{k}} M^{\frac{1}{2}} + \|u_0\|_{2,k} \right),$$

for a constant $C = C(\Omega, k, \lambda, A)$. This gives the equi-coercivity of $(E_p)_p$ in $W^{2,k}_{u_0}$.
Moreover, $(E_p)_p$ is a monotone increasing sequence of weakly lower semicontinuous functionals on $W^{2,k}_{u_0}$, thus its Γ-limit is given by (see [6, Remark 2.12])

$$\Gamma - \lim_p E_p = \lim_p E_p = E_\infty.$$

By the Fundamental Theorem of Gamma Convergence [6, Thm. 2.10], if $(u_p)_p$ is a sequence of minimisers of E_p, up to passing to a subsequence, we have $u_p \rightharpoonup u_\infty$ weakly $W^{2,k}_{u_0}$ where u_∞ is a minimiser of E_∞ on $W^{2,k}_{u_0}$. By definition of E_∞, we necessarily have $u_\infty \in \mathcal{W}^{2,\infty}_{u_0}(\Omega, \mathbb{R}^N)$. Moreover, by the same Theorem

$$E_p(u_p) \to E_\infty(u_\infty) \quad \text{as } p \to \infty. \tag{3.2}$$

3.2 PDE derivation

Now we work towards the proof of (**??**). Let us begin by writing down the system of Euler-Lagrange equations satisfied by u_p, using the variational nature of the operator L and the symmetry of A:

$$\mathrm{L}f_p = 0, \quad \text{in } \mathcal{D}'(\Omega, \mathbb{R}^N), \tag{3.3}$$

where, for $e_p := E_p(u_p)$, $f_p : \Omega \to \mathbb{R}^N$ is defined by

$$f_p := e_p^{1-p} F(\cdot, \mathrm{L}u_p)^{p-1} F_\xi(\cdot, \mathrm{L}u_p). \tag{3.4}$$

Since the only interesting case is $e_\infty > 0$, we may assume that $e_p \geq \alpha > 0$ (for p large enough). In fact, if $e_\infty = 0$, then $F(\cdot, \mathrm{L}u_\infty) = 0$ and system (**??**) reduces to $\mathrm{L}u_\infty = 0$. For this reason, f_p is a well defined measurable map.
Let us prove that that $(f_p)_p$ is bounded in $L^1(\Omega, \mathbb{R}^N)$:

$$\begin{aligned}
\fint_\Omega |f_p(x)|dx &= e_p^{1-p} \fint_\Omega F(x, \mathrm{L}u_p(x))^{p-1} |F_\xi(x, \mathrm{L}u_p(x))| dx \\
&\leq c e_p^{1-p} \fint_\Omega F(x, \mathrm{L}u_p(x))^{p-1} |\mathrm{L}u_p(x)| dx \\
&\leq c e_p^{1-p} \left(\fint_\Omega F(x, \mathrm{L}u_p(x))^p dx \right)^{\frac{p-1}{p}} \left(\fint_\Omega |\mathrm{L}u_p(x)|^p dx \right)^{\frac{1}{p}} \\
&= c \left(\fint_\Omega |\mathrm{L}u_p(x)|^p dx \right)^{\frac{1}{p}} \\
&\leq c \left(\fint_\Omega [cF(x, \mathrm{L}u_p(x))]^{\frac{p}{2}} dx \right)^{\frac{1}{p}} \\
&\leq c c^{\frac{1}{2}} \left(\fint_\Omega F(x, \mathrm{L}u_p(x))^p dx \right)^{\frac{1}{2p}} = c^{\frac{3}{2}} e_p^{\frac{1}{2}} \leq c^{\frac{3}{2}} e_\infty^{\frac{1}{2}}.
\end{aligned} \tag{3.5}$$

In particular, thanks to Proposition **??**, $f_p \in W^{1,2}_{loc}(\Omega, \mathbb{R}^N) \cap L^1(\Omega, \mathbb{R}^N)$ is a weak solution of (3.3). Thus, by Corollary **??**, for every $q < \infty$,

$$\|f_p\|_{2,q,U} \leq C \|f_p\|_1 \leq C, \quad \forall U \Subset \Omega,$$

for some constant $C = C(q, U, \Omega, c, e_\infty) > 0$. Hence, by Morrey's theorem, we deduce that $(f_p)_p$ is bounded in the local topology of C^1, and so there exists f_∞ and a subsequence $(f_{p_\ell})_\ell$ such that $f_{p_\ell} \to f_\infty$ locally uniformly as $\ell \to \infty$. In particular, $\mathrm{L}f_\infty = 0$ in the sense of distributions and $f_\infty \in L^1(\Omega, \mathbb{R}^N)$ as well.
The next lemma is central in our argument.

Lemma 3.1. (Non-triviality of f_∞) *The map f_∞ constructed above is such that $f_\infty \not\equiv 0$.*

Proof. First we show that $f_p \overset{*}{\rightharpoonup} f_\infty$ in $\mathcal{M}(\overline{\Omega}, \mathbb{R}^N)$, as $p \to \infty$ along a subsequence.
By approximating with C^∞_c-maps, thanks to the regularity assumptions on A, for every $p \in (1, \infty)$ we have

$$\mathrm{L}f_p = \mathrm{L}f_\infty = 0 \quad \text{in } \left(W^{2,\infty}_0(\Omega, \mathbb{R}^N) \right)^*. \tag{3.6}$$

From the boundedness in L^1, there exists a measure $\mu \in \mathcal{M}(\overline{\Omega}, \mathbb{R}^N)$ such that $f_p \mathscr{L}^n \llcorner \Omega \overset{*}{\rightharpoonup} \mu$ in $\mathcal{M}(\overline{\Omega}, \mathbb{R}^N)$. Since $f_p \rightharpoonup f_\infty$ in $W^{2,q}_{loc}(\Omega, \mathbb{R}^N)$, we necessarily have $\mu \llcorner \Omega = f_\infty \mathscr{L}^n \llcorner \Omega$. So we can write

$$\mu = f_\infty \mathscr{L}^n \llcorner \Omega + \mu \llcorner \partial\Omega.$$

Now we claim that $\mu \llcorner \partial\Omega = 0$. Let us denote $d := \mathrm{dist}(\cdot, \partial\Omega)$ and $\Omega_r := \{x \in \Omega : d(x) \leq r\}$. Then, by the regularity of $\partial\Omega$, we have $d \in C^2(\overline{\Omega}_r)$ for some $r > 0$ sufficiently small. Extend it to a function $d \in C^2(\overline{\Omega})$ and fix a map $g \in C^2(\overline{\Omega}, \mathbb{R}^N)$. Set $\psi := d^2 g \in C^2(\overline{\Omega}, \mathbb{R}^N) \cap W^{2,\infty}_0(\Omega, \mathbb{R}^N)$ and test (3.6) with ψ:

$$0 = \int_\Omega (f_p - f_\infty) \cdot \mathrm{L}\psi = \int_\Omega (f_p - f_\infty) \cdot [(\mathrm{div} A)\mathrm{D}\psi + A\mathrm{D}^2\psi]. \tag{3.7}$$

By a computation,

$$\begin{aligned} \mathrm{D}\psi &= 2d\, g \otimes \mathrm{D}d + d^2\mathrm{D}g, \\ \mathrm{D}^2\psi &= 2g \otimes \mathrm{D}d \otimes \mathrm{D}d + 2d\, g \otimes \mathrm{D}^2 d + 4d\mathrm{D}g \otimes \mathrm{D}d + d^2\mathrm{D}^2 g. \end{aligned}$$

Thus,

$$\mathrm{D}^2\psi \llcorner \partial\Omega = 2(g \otimes \mathrm{D}d \otimes \mathrm{D}d) \llcorner \partial\Omega = 2\, g \otimes \nu \otimes \nu \llcorner \partial\Omega,$$

where ν is the outwards unit normal to the boundary. Since $(\mathrm{div}A)\mathrm{D}\psi \in C_0(\Omega, \mathbb{R}^N)$ and $(f_p - f_\infty)\mathscr{L}^n \llcorner \Omega \overset{*}{\rightharpoonup} \mu \llcorner \partial\Omega$ in $\mathcal{M}(\overline{\Omega}, \mathbb{R}^N)$, we can pass to the limit in (3.7) to obtain

$$0 = 2\int_{\partial\Omega} A : g \otimes \nu \otimes \nu \cdot d\mu.$$

By the ellipticity of A, the matrix $M := A : \nu \otimes \nu$ is positive definite, so we conclude that

$$\int_{\partial\Omega} Mg \cdot d\mu = 0, \quad \forall g \in C^2(\overline{\Omega}, \mathbb{R}^N),$$

which gives $\mu \llcorner \partial\Omega = 0$.
By Lemma 2.3, we estimate

$$\begin{aligned} \int_\Omega f_p \cdot \mathrm{L}u_p &= e_p^{1-p} \int_\Omega F(x, \mathrm{L}u_p(x))^{p-1} F_\xi(x, \mathrm{L}u_p(x)) \cdot \mathrm{L}u_p(x) dx \\ &\geq e_p^{1-p} \int_\Omega F(x, \mathrm{L}u_p(x))^p dx \\ &= e_p^{1-p} e_p^p |\Omega| = |\Omega| e_p \geq |\Omega|\alpha, \end{aligned}$$

where in the last step, we recall we are under the assumption $e_\infty > 0$.
Thanks to Whitney's extension theorem, there exists $u_* \in C^2(\mathbb{R}^n, \mathbb{R}^N)$ such that $u_* = u_0$ and $\mathrm{D}u_* = \mathrm{D}u_0$ on $\partial\Omega$. Now we observe that (3.3) clearly holds also in $\left(W^{2,2p}_0(\Omega, \mathbb{R}^N)\right)^*$, therefore by testing with $u_p - u_* \in W^{2,2p}_0(\Omega, \mathbb{R}^N)$, we obtain

$$|\Omega|\alpha \leq \int_\Omega f_p \cdot \mathrm{L}u_p = \int_\Omega f_p \cdot \mathrm{L}u_*.$$

So, we can pass to the limit on the right hand side, observing that $\mathrm{L}u_* \in C(\overline{\Omega}, \mathbb{R}^N)$, and conclude

$$\int_\Omega f_\infty \cdot \mathrm{L}u_* \geq |\Omega|\alpha.$$

In particular, this ensures that $f_\infty \not\equiv 0$. $\square$

Thanks to Lemma 3.1, we are now able to derive the PDE satisfied by u_∞.
Set $\Gamma_\infty := f_\infty^{-1}(0) \subsetneq \Omega$. From (3.4), we obtain the equation

$$|f_p(x)|^{\frac{2}{2p-1}} \frac{f_p(x)}{|f_p(x)|} = e_p^{-1+\frac{1}{2p-1}} F(x, \mathrm{L}u_p(x)) \left(\frac{|F_\xi(x, \mathrm{L}u_p(x))|}{\sqrt{F}(x, \mathrm{L}u_p(x))} \right)^{\frac{2}{2p-1}} \frac{F_\xi(x, \mathrm{L}u_p(x))}{|F_\xi(x, \mathrm{L}u_p(x))|}, \qquad \text{a.e. } x \in \Omega, \tag{3.8}$$

which we rewrite as

$$e_p^{1-\frac{1}{2p-1}}\left(\frac{|F_\xi(x,\mathrm{L}u_p(x))|}{\sqrt{F}(x,\mathrm{L}u_p(x))}\right)^{-\frac{2}{2p-1}}|f_p(x)|^{\frac{2}{2p-1}}\frac{f_p(x)}{|f_p(x)|}=\Phi(x,\mathrm{L}u_p(x)),\qquad \text{a.e. } x\in\Omega, \tag{3.9}$$

where we have symbolised

$$\Phi(x,\xi):=F(x,\xi)\frac{F_\xi(x,\xi)}{|F_\xi(x,\xi)|},\qquad \text{a.e. } x\in\Omega,\quad \forall\xi\in\mathbb{R}^N.$$

Notice that equations (3.8) and (3.9) are well defined. Indeed, from (**??**),(**??**), and Lemma 2.3, it is easy to see that $|F_\xi|/\sqrt{F}$ is bounded from above and below. Moreover, by the assumptions on F, we have $f_p=0$ on a set of positive measure $E\subset\Omega$ if and only if $\mathrm{L}u_p=0$ on the same set. So, points in E are not singular for both sides of (3.8) and (3.9), naturally vanishing on E.
Now we recall that by Lemma 2.4, for almost every $x\in\Omega$, the map $\Phi_x:=\Phi(x,\cdot):\mathbb{R}^N\to\mathbb{R}^N$ is a homeomorphism. So, applying the inverse map Φ_x^{-1}, we rewrite the previous equation as

$$\mathrm{L}u_p(x)=\Phi_x^{-1}\left(e_p^{1-\frac{1}{2p-1}}\left(\frac{|F_\xi(x,\mathrm{L}u_p(x))|}{\sqrt{F}(x,\mathrm{L}u_p(x))}\right)^{-\frac{2}{2p-1}}|f_p(x)|^{\frac{2}{2p-1}}\frac{f_p(x)}{|f_p(x)|}\right),\qquad \text{a.e. } x\in\Omega.$$

Recall that $f_{p_\ell}\to f_\infty$ uniformly on every compact $K\subset\Omega\setminus\Gamma_\infty$ and, by (3.2), $e_{p_\ell}\to e_\infty$ as $\ell\to\infty$. So we can pass to the limit locally uniformly in the right hand side along the subsequence $(p_\ell)_\ell$. On the other hand, since we already know that $\mathrm{L}u_p \rightharpoonup \mathrm{L}u_\infty$ weakly in L^q for every $q\geq 1$, we obtain

$$\mathrm{L}u_\infty(x)=\Phi_x^{-1}\left(e_\infty\frac{f_\infty(x)}{|f_\infty(x)|}\right),\qquad \text{a.e. } x\in K.$$

By inverting again, we get

$$\Phi(x,\mathrm{L}u_\infty)=F(x,\mathrm{L}u_\infty)\frac{F_\xi(x,\mathrm{L}u_\infty)}{|F_\xi(x,\mathrm{L}u_\infty)|}=e_\infty\frac{f_\infty(x)}{|f_\infty(x)|},\qquad \text{a.e. } x\in K.$$

Since $\mathrm{L}f_\infty=0$, the unique continuation property of L yields $|\Gamma_\infty|=0$. As a consequence, we obtain (**??**).

3.3 Uniqueness

In the previous sections, we proved the existence of a minimiser $u_\infty\in\mathcal{W}^{2,\infty}_{u_0}(\Omega,\mathbb{R}^N)$ of E_∞ and that it satisfies the PDE system (**??**). As a first step to provide uniqueness of minimisers of E_∞ in $\mathcal{W}^{2,\infty}_{u_0}(\Omega,\mathbb{R}^N)$, we show that every minimiser of E_∞ satisfies a PDE system of the same structure as (**??**). Some of the arguments are similar to Section 3.1, but we will present the complete proof for sake of clarity. More precisely, we start with proving the following result.

Lemma 3.2. (Necessity of the PDE system) *Let F, A, and u_0 be as in Theorem **??**. Suppose that $u\in\mathcal{W}^{2,\infty}_{u_0}(\Omega,\mathbb{R}^N)$ is a minimiser of E_∞. Then there exists $f\in L^1(\Omega,\mathbb{R}^N)$ such that $f\in W^{2,q}_{loc}(\Omega,\mathbb{R}^N)$ for all $q<\infty$ and*

$$\begin{cases}F(x,\mathrm{L}u(x))\dfrac{F_\xi(x,\mathrm{L}u(x))}{|F_\xi(x,\mathrm{L}u(x))|}=e_\infty\dfrac{f(x)}{|f(x)|}, & a.e.\ x\in\Omega,\\ \mathrm{L}f(x)=0, \quad a.e.\ x\in\Omega.\end{cases} \tag{3.10}$$

Proof. Let $u \in \mathcal{W}^{2,\infty}_{u_0}(\Omega, \mathbb{R}^N)$ be a minimiser of E_∞. For any $p \geq 1$, we consider the auxiliary functional

$$A_p(v) := E_p(v) + \frac{1}{2} \fint_\Omega |v-u|^2, \qquad v \in W^{2,2p}_{u_0}(\Omega, \mathbb{R}^N).$$

Using the Direct Method, it is not difficult to see that A_p attains a minimum point. Let $v_p \in W^{2,2p}_{u_0}(\Omega, \mathbb{R}^N)$ be a minimiser of A_p, then, recalling that u_p is a minimiser of E_p in the same space, we have

$$e_p = E_p(u_p) \leq E_p(v_p) \leq A_p(v_p) \leq A_p(u) = E_p(u) \leq E_\infty(u) = e_\infty.$$

Since $e_p \to e_\infty$, we have $A_p(v_p) \to e_\infty$ as $p \to \infty$. Now, if $(v_p)_p$ is a sequence of minimisers of A_p, by Calderon-Zygmund estimates, we can prove as in Section 3.1 that $(v_p)_p$ bounded in $W^{2,q}_{u_0}(\Omega, \mathbb{R}^N)$ for any fixed $q \geq 1$. So, $v_{p_\ell} \rightharpoonup v_\infty$ weakly $W^{2,q}_{u_0}$ for a subsequence $(p_\ell)_\ell$ and some $v_\infty \in W^{2,q}_{u_0}(\Omega, \mathbb{R}^N)$. We claim that $v_\infty = u$. Indeed, passing to the limit as $\ell \to \infty$ in the following inequality

$$e_{p_\ell} + \frac{1}{2} \fint_\Omega |v_{p_\ell} - u|^2 \leq A_{p_\ell}(v_{p_\ell}),$$

we get

$$e_\infty + \frac{1}{2} \fint_\Omega |v_\infty - u|^2 \leq e_\infty,$$

which necessarily implies $v_\infty = u$.
Now, for simplicity let us denote $v_p = v_{p_\ell}$ and write down the Euler-Lagrange equation satisfied by v_p:

$$\mathrm{L} g_p + v_p - u = 0 \qquad \text{in } \mathcal{D}'(\Omega, \mathbb{R}^N), \tag{3.11}$$

where, for $a_p := E_p(v_p)$, g_p is defined by $g_p := a_p^{1-p} F(\cdot, \mathrm{L}v_p)^{p-1} F_\xi(\cdot, \mathrm{L}v_p)$. We can prove that g_p is well defined and bounded in L^1 as showed for f_p in Section 3.1. In particular, thanks to Proposition **??**, $g_p \in W^{1,2}_{loc}(\Omega, \mathbb{R}^N) \cap L^1(\Omega, \mathbb{R}^N)$ is a weak solution of (3.11). Thus, by Corollary **??**, for every $q < \infty$,

$$\|g_p\|_{2,q,U} \leq C\left(\|g_p\|_1 + \|v_p - u\|_q\right) \leq C, \quad \forall U \Subset \Omega,$$

for some constant $C = C(q, U, \Omega, c, e_\infty) > 0$. By Sobolev embeddings, up to passing to a (not relabeled) subsequence, $g_p \to g_\infty$ locally uniformly as $p \to \infty$, for some $g_\infty \in L^1(\Omega, \mathbb{R}^N) \cap W^{2,q}_{loc}(\Omega, \mathbb{R}^N)$ such that $\mathrm{L}g_\infty = 0$. We can prove that $g_\infty \not\equiv 0$ as in Lemma 3.1, since the same argument holds even if $\mathrm{L}g_p \rightharpoonup 0$ in the sense of distributions only. Finally, since $a_p \to e_\infty$ as $p \to \infty$, we can pass to the limit in the expression defining g_p exactly as we did for f_p, obtaining (3.10) with $f = g_\infty$. ☐

We conclude this section with the proof of uniqueness of the solution to the minimum problem associated to E_∞. Let u_1 and u_2 be two minimisers in $\mathcal{W}^{2,\infty}_{u_0}(\Omega, \mathbb{R}^N)$, then, by (3.10), we have

$$F(\cdot, \mathrm{L}u_1) = F(\cdot, \mathrm{L}u_2) = e_\infty, \quad \text{a.e. in } \Omega.$$

By convexity of $F(x, \cdot)$ and linearity of L, for a.e. $x \in \Omega$ we have

$$\begin{aligned} F\left(x, \mathrm{L}\left(\frac{u_1(x) + u_2(x)}{2}\right)\right) &= F\left(x, \frac{\mathrm{L}u_1(x)}{2} + \frac{\mathrm{L}u_2(x)}{2}\right) \\ &\leq \frac{1}{2} F(x, \mathrm{L}u_1(x)) + \frac{1}{2} F(x, \mathrm{L}u_2(x)) \\ &= e_\infty. \end{aligned} \tag{3.12}$$

The average $\frac{u_1+u_2}{2}$ still belongs to $\mathcal{W}^{2,\infty}_{u_0}(\Omega,\mathbb{R}^N)$, then necessarily

$$F\left(\cdot\,,\mathrm{L}\left(\frac{u_1+u_2}{2}\right)\right) = e_\infty, \quad \text{a.e. in } \Omega.$$

But $F(x,\cdot)$ is strictly convex, so the only possibility to have equality in (3.12) is that $\mathrm{L}u_1 = \mathrm{L}u_2$, giving $u_1 = u_2$ by uniqueness results for elliptic systems with Dirichlet boundary conditions (see [14, Theorems 3.39 and 3.46]).

4 On the unique continuation property

In this last section, we identify classes of operators satisfying the unique continuation property (see Definition **??**). First, it is worth mentioning that a slightly different notion that often appears in the literature is the *strong unique continuation property*, namely the only solution to $\mathrm{L}u = 0$ with a zero of infinite order is the trivial one. We recall that $x_0 \in \Omega$ is a zero of infinite order for u if

$$\int_{B_r(x_0)} |u|^2 dx = O(r^k) \qquad \text{as } r \to 0 \quad \forall k \in \mathbb{N}.$$

There is also the definition of *weak unique continuation property* (the only solution to $\mathrm{L}u = 0$ vanishing on an open set is the trivial one), but we are not interested in that one. For a proof of the fact that the strong definition implies the one in Definition **??**, we refer to [15]. Their proof actually applies to elliptic equations, but it can be identically readapted for systems, since it is basically a consequence of Caccioppoli inequalities [14, Sec. 4.2]. However, differently from the scalar case, where the zero nodal set is even $(n-1)$-dimensional [16], we have no theory available that guarantees sufficient conditions for (strong) unique continuation property for elliptic systems. For this reason, we restrict ourself to list some examples of interest, although one of them is deeply connected to the scalar case. In what follows, we assume that $u \in L^1(\Omega,\mathbb{R}^N)$ is a distributional solution of $\mathrm{L}u = \mathrm{div}(A\mathrm{D}u) = 0$ with A satisfying either (H1) or (H2). Notice that by Proposition **??**, u is actually a weak solution. Without loss of generality, we assume that u is not constant. We consider the following cases:

- A has analytic coefficients.
  In this case, it is well known that solutions to $\mathrm{L}u = 0$ are analytic [36]. In particular, if u is not constant, its nodal sets are negligible.

- A has the form
  $$A = \sum_{h=1}^{N} e_h \otimes e_h \otimes B_h$$
  with $\{e_h\}_{h=1,\dots,N}$ orthonormal base of $\mathbb{R}^N$ and $B_h \in C^1(\overline{\Omega},\mathbb{R}^{n\times n})$ elliptic for every $h = 1,\dots,N$.
  We have
  $$(\mathrm{L}u)_i = \mathrm{D}_\alpha\left(\left[\sum_{h=1}^{N} e_{h_i} e_{h_j} B_h^{\alpha\beta}\right] \mathrm{D}_\beta u^j\right) = 0 \quad \forall i = 1,\dots,N.$$

Fix $k \in \{1, \dots, N\}$ and take the scalar product of Lu with e_k:

$$
\begin{aligned}
0 &= e_k^i (\mathrm{L}u)_i \\
&= \mathrm{D}_\alpha \left(\left[\sum_{h=1}^N e_k^i e_{h_i} e_{h_j} B_h^{\alpha\beta} \right] \mathrm{D}_\beta u^j \right) \\
&= \mathrm{D}_\alpha \left(e_{k_j} B_k^{\alpha\beta} \mathrm{D}_\beta u^j \right) \\
&= \mathrm{D}_\alpha \left(B_k^{\alpha\beta} \mathrm{D}_\beta (e_{k_j} u^j) \right)
\end{aligned}
$$

Set $u^{(k)} = e_k \cdot u$, namely the k-th component of u w.r.t. the base $\{e_h\}$. Then, for each $k = 1, \dots, N$, $u^{(k)}$ solves an elliptic equation in divergence form. Since we assumed that u is not constant, there is at least one k such that $u^{(k)}$ is not constant. So, by applying the scalar theory [16] to $u^{(k)}$, since $\{|u| = 0\} \subset \{u^{(k)} = 0\}$, the unique continuation must hold.

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